Department of Computer Engineering

CHESTXPLAIN

Multi-Label Chest X-Ray Disease Classification and Explainability with Deep Learning

Abstract

ChestXplain classifies 14 thoracic diseases from X-rays using deep learning. With Grad-CAM explainability and a native mobile app, it provides clinicians with real-time, transparent diagnostic support.

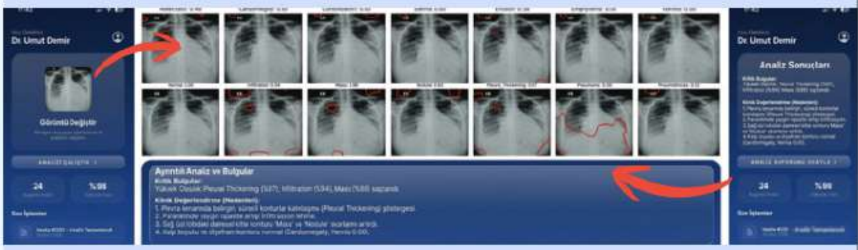
Introduction

- Challenge of accurately detecting multiple concurrent thoracic diseases from standard chest X-rays.
- Shortage of expert radiologists.
- Importance of explainable AI (XAI) in critical medical applications to build trust and understanding of model decisions.

Solution

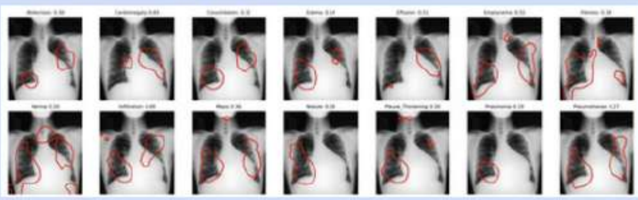
- Chest X-rays are processed by a DenseNet-121 backbone integrated with CBAM for enhanced feature extraction.
- Generates on-demand Grad-CAM heatmaps to visualize the model's exact focal points.
- High-performance FastAPI backend paired with a secure MongoDB database.
- Diagnostic insights and visual reasoning are delivered seamlessly through a native mobile application.

System Pipeline & Mobile App



An end-to-end Android and FastAPI system using DenseNet-121 for real-time multi-label predictions and Grad-CAM explainability.

EXPLAINABLE AI: GRAD-CAM VISUALIZATION



Heatmap overlay highlights the specific pathological regions driving the deep learning model's multi-label prediction, ensuring diagnostic transparency.

Results & Methodology

- Architecture: DenseNet-121 backbone enhanced with Convolutional Block Attention Modules (CBAM) to focus on subtle radiographic patterns.
- Class Imbalance: Addressed severe dataset imbalance using Asymmetric Loss (ASL) and a weighted learning strategy.
- Performance: Evaluated using F1 and Macro F1 scores to ensure robust detection across 14 different thoracic conditions.

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